

Joseph Green

Senior Data Engineer / Data Architect (Python | AWS | Databricks | Snowflake)

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SUMMARY

Senior Data Engineer & Data Architect with 10+ years of experience designing **large-scale data platforms**, distributed data pipelines, and real-time analytics systems across **finance**, **SaaS**, and **healthcare**. Expert in **AWS**, **Python**, Java, SQL, Databricks, dbt, Snowflake, Airflow, Kafka, Terraform, Docker, and Kubernetes. Proven ability to architect high-throughput **ETL/ELT** workflows, build reliable **data ingestion / transformation frameworks**, and deploy production-grade pipelines supporting millions of daily events. Known for improving pipeline reliability, observability, and performance while collaborating closely with engineering, analytics, and product teams to deliver scalable and compliant data platforms.

TECHNICAL SKILLS

- Languages: Python, SQL, Java, Scala, Bash
- Frameworks: Apache Spark, PySpark, Spring Boot, Flask, FastAPI
- Pipelines & Orchestration: Airflow, dbt, Databricks, Kafka, Kinesis, Flink, EMR
- Cloud (Primary): AWS (S3, EMR, Glue, Lambda, RDS, DynamoDB, ECS, EKS, Athena, Redshift)
- Data Warehouses: Snowflake, ClickHouse, Delta Lake, Iceberg, Hive
- Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis
- Infrastructure & DevOps: Docker, Kubernetes (EKS), Terraform, Helm, GitHub Actions, Jenkins
- Observability: Datadog, CloudWatch, Prometheus, OpenTelemetry
- Data Tools: Airbyte, Rudderstack, Great Expectations, Apache Atlas
- Architecture: Data Lakehouse, ETL/ELT, distributed systems, event-driven design, real-time analytics

EDUCATION

UNIVERSITY OF TEXAS AT DALLAS	Dallas, TX
<i>Master of Science, Data Science</i>	2015-2016

UNIVERSITY OF TEXAS AT DALLAS	Dallas, TX
<i>Bachelor of Science, Information Technology</i>	2011-2015

PROFESSIONAL EXPERIENCE

JP MORGAN CHASE & CO	Plano, TX
<i>Senior Data Engineer Data Architect</i>	01/2023-Present

- Served as the lead Data Engineer & Data Architect for JP Morgan's financial data platform, owning design and delivery of large-scale data pipelines, real-time streaming frameworks, ETL/ELT workflows, observability strategy, and metadata governance across AWS.
- Built production systems using Python, Java, SQL, Spark, Databricks, Airflow, dbt, Kafka, Delta Lake, ClickHouse, Terraform, Docker, and EKS to support high-volume financial data processing, fraud detection, and regulatory reporting.
- Improved processing speed, reliability, and platform scalability across millions of daily transactions.

- Architected high-throughput **Spark** and **PySpark** pipelines on **Databricks**, **EMR**, and **Glue**, processing 25M+ transactions/day with optimized cluster autoscaling and partitioning strategies.
- Designed a **Delta Lake architecture** on AWS S3 with ACID transactions, schema evolution, time travel, and data versioning that reduced data corruption incidents by 35%.
- Built 300+ production **Airflow DAGs** with SLA tracking, data quality triggers, retry logic, and lineage propagation—improving pipeline uptime and reliability by 40%.
- Developed **real-time streaming pipelines** using **Kafka + Spark Structured Streaming** to support sub-second fraud detection and event-driven risk signals for 5,000+ internal users.
- Designed and implemented **dbt models** for financial transformations, dimensional modeling, and semantic layer standardization for analytics and reporting teams.
- Integrated **ClickHouse** as a high-performance OLAP engine, enabling real-time analytics and sub-second query responses for operational dashboards.
- Built **Terraform IaC modules** for provisioning EMR clusters, MSK topics, S3 buckets, IAM roles, and VPC networking, ensuring reproducible and secure cloud deployments.
- Containerized ETL workloads using **Docker**, orchestrated distributed workloads using **Kubernetes (EKS)**, and implemented Helm-based deployment pipelines.
- Implemented advanced data validation frameworks using Great Expectations, enforcing schema rules, anomaly thresholds, and audit-ready quality checks.
- Deployed robust **monitoring** and **observability** with Datadog and CloudWatch, enabling deep visibility into pipeline latency, job failures, cluster performance, and throughput.
- Engineered **real-time ingestion frameworks** using Airbyte + Kafka + Lambda for ingesting structured and semi-structured market and trading data.
- Collaborated with finance, compliance, fraud, and reporting teams to translate regulatory requirements into enforceable **data governance rules**, **lineage policies**, and **access controls**.
- Led cross-team architectural reviews, mentored engineers, and established coding standards for distributed ETL systems, enabling consistent engineering patterns across the organization.

BLUECORE INNOVATIONS

Dallas, TX

Senior Data Engineer

05/2018-12/2022

- Owned the design, build, and optimization of ETL pipelines, event-driven data services, and Snowflake-based modeling layers for a SaaS marketing automation platform.
- Delivered scalable solutions using Python, SQL, dbt, Airflow, Kafka, Spark, Snowflake, Docker, Terraform, and **AWS** to support high-volume customer analytics, personalization pipelines, and cross-channel marketing workflows.
- Orchestrated 50+ ETL/ELT pipelines supporting customer behavior analytics, real-time events, and SaaS product telemetry with strong reliability and automated validation.
- Built modular **Airflow DAGs** that replaced legacy SSIS workloads, reducing operational overhead by 40% and improving monitoring consistency.
- Designed, maintained, and optimized **Snowflake schemas** using clustering keys, materialized views, and warehouse tuning to reduce report generation time by 50%.

- Managed **Kafka** clusters for event streaming between CRM, marketing, and engagement systems with uptime exceeding 99.9%.
- Developed **dbt transformations** with reusable macros, staged models, and production-grade semantic layers aligned to marketing KPIs.
- Implemented **PostgreSQL performance optimization** using indexing strategies, table partitioning, and vacuum tuning to enhance ETL staging performance.
- Automated CI/CD pipelines using **Jenkins** and **GitHub Actions**, supporting environment-specific builds, dbt test enforcement, and Airflow deployment automation.
- Used **Datadog observability** to instrument pipeline metrics, alerting, and distributed tracing for ETL and streaming systems.
- Implemented secure access controls, field-level masking, and data governance using Snowflake + Apache Atlas to confirm compliance with customer data privacy requirements.
- Partnered with product, analytics, and data science teams to translate SaaS feature requirements into scalable ingestion and modeling workflows powering personalization, segmentation, and targeting.

NEXTTECH SOLUTIONS

Austin, TX

Data Engineer

12/2016-04/2018

- Engineered **healthcare**-grade ETL pipelines and secure data processing frameworks for an EHR and insurance analytics platform, integrating claims data, medical records, and patient outcomes.
- Delivered production systems using Python, SQL, Flask APIs, Docker, Airflow, PostgreSQL, AWS, and Great Expectations to support **HIPAA**-compliant workflows and analytics dashboards.
- Developed ETL workflows for processing 10M+ healthcare records monthly, ensuring HIPAA compliance and strong access controls.
- Designed PostgreSQL schemas for EHR and claims data with optimized indexes, table partitioning, and analytical query tuning that reduced processing time by 40%.
- Built secure **Flask APIs** for ingesting patient data from provider systems, handling 5,000+ requests daily with authentication and audit logging.
- Containerized ETL operations using **Docker** and created reproducible runtime environments for analytics and data science teams.
- Automated recurring reporting and validation using **Airflow DAGs** with data quality scripts to ensure 98%+ accuracy across critical metrics.
- Implemented **AWS S3 versioning** and **cross-region replication** for data durability, reducing recovery point objectives to under one hour.
- Built 15+ healthcare analytics dashboards using **Tableau**, integrating clinical, claims, and operational metrics for stakeholders and care teams.